European Journal of Surgical Oncology 50 (2024) 108732

Contents lists available at ScienceDirect

European Journal of Surgical Oncology

journal homepage: www.ejso.com

Predictors of mastectomy in breast cancer patients with complete remission of primary tumor after neoadjuvant therapy: A retrospective study

Damiano Gentile [a,b,1], Federica Martorana [c,d,*,1], Andreas Karakatsanis [e,f], Francesco Caruso [d], Michele Caruso [d], Gaetano Castiglione [d], Alfio Di Grazia [d], Francesco Pane [d], Antonio Rizzo [d], Paolo Vigneri [c,d], Corrado Tinterri [a,b], Giuseppe Catanuto [d,g]

[a] Breast Unit, IRCCS Humanitas Research Hospital, Rozzano, Milan, Italy
[b] Department of Biomedical Sciences, Humanitas University, Pieve Emanuele, Milan, Italy
[c] University of Catania, Department of Clinical and Experimental Medicine, Catania, Italy
[d] Humanitas Istituto Clinico Catanese, Misterbianco, Catania, Italy
[e] Department for Surgical Sciences, Uppsala University, Uppsala, Sweden
[f] Section for Breast Surgery, Department of Surgery, Uppsala University Hospital, Uppsala, Sweden
[g] G.Re.T.A. Group for Reconstructive and Therapeutic Advancements Fondazione ETS, Naples, Italy

ARTICLE INFO

Keywords:
Breast cancer
Neo-adjuvant therapy
Breast-conserving surgery
Mastectomy
Decision-making

ABSTRACT

Introduction: Neoadjuvant therapy (NAT) should increase the rate of breast-conserving surgery (BCS) in non-metastatic breast cancer (BC) patients, especially in those achieving tumor shrinkage. Still, the conversion from a pre-planned mastectomy to BCS in patients responding to NAT is not a widespread standard. We aimed to identify factors influencing surgical choices in this setting.

Materials and methods: We retrospectively collected data of BC patients with complete remission of primitive tumor (ypT0) after NAT, treated with BCS or mastectomy in two Italian breast units. Predictors of mastectomy were explored using logistic regression. Distant recurrence and event-free survival were assessed in the BCS and mastectomy cohort.

Results: 243 patients were included, 147 (60.5 %) treated with BCS and 96 (39.5 %) treated with mastectomy. In the mastectomy group, there were more centrally-located, multiple and larger tumors. At univariate regression analysis, central location, baseline tumor extension on ultrasound (US) and magnetic resonance imaging (MRI), multiple foci and clinical stage were significantly associated with the chance of receiving mastectomy. At multivariate analysis, only baseline focality on US and extension on MRI retained significance as predictors of mastectomy. Distant recurrence and event-free survival were significantly longer in patients undergoing BCS.

Conclusion: Baseline tumor extension and focality were the main predictors of mastectomy in patients with ypT0 after NAT. However, BCS did not negatively affect survival outcomes in our cohort. An effort should be made to avoid potentially unnecessary mastectomy in this population, aiming at minimizing surgery-associated toxicities and improving patients' quality of life.

1. Introduction

Neo-adjuvant therapy (NAT) has gained increasing importance in the management of non-metastatic breast cancer (BC) patients. Initially exploited to convert inoperable to operable BC [1–3], NAT allows *in-vivo* assessment of tumor chemo-sensitivity, guiding therapeutic choices in the post-operative setting [4,5]. The degree of response to NAT exerts prognostic value, significantly related to survival outcomes [4,6–9]. Lastly, achieving tumor shrinkage can reduce the extent of surgery, avoiding unnecessary axillary lymph node dissection [10,11] and enabling breast-conserving surgery (BCS) in patients initially scheduled for mastectomy [12–16]. It is widely recognized that BCS after NAT is feasible and safe [17–19]. Still, the conversion from a pre-planned mastectomy to BCS in patients responding to NAT is not a universal standard [20], despite the advances in imaging techniques and in localization of non-palpable breast lesions.

* Corresponding author. Department of Clinical and Experimental Medicine, University of Catania, Catania, 95123, Italy.
E-mail address: federica.martorana@unict.it (F. Martorana).
[1] These authors equally contributed to this work.

https://doi.org/10.1016/j.ejso.2024.108732
Received 9 June 2024; Received in revised form 20 September 2024; Accepted 30 September 2024
Available online 2 October 2024
0748-7983/© 2024 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

The choice between BCS and mastectomy often remains a complex and multifaceted decision for breast surgeons. This decision can be influenced by several patient, tumor, and histopathological features. Understanding these factors is crucial for optimizing treatment strategies and aligning surgical decisions with the latest clinical evidence.

We retrospectively collected and analyzed data from BC patients treated in two Italian Breast Units who achieved a complete response on the primary tumor (*i.e.*ypT0) after NAT and were subsequently treated with BCS or mastectomy. The aim of this study is to identify factors influencing surgical decision-making and predicting the use of mastectomy despite the tumor's complete response to NAT.

2. Materials and Methods

2.1. Study design and participants

This study is a pooled analysis from DataBreast, a prospectively maintained European Society of BC Specialist (EUSOMA)-approved institutional database. Patients informed consent was obtained before performing data entry.

We collected data from all consecutive BC patients treated with NAT followed by surgery at the Breast Unit of Humanitas Istituto Clinico Catanese (HICC, Catania, Italy), and IRCCS Humanitas Research Hospital (ICH, Rozzano, Italy) between 2009 and 2022. Only patients with a documented pathologic complete response of the primary tumor (ypT0), defined as the absence of residual invasive or *in-situ* carcinoma on the surgical specimen obtained after NAT were eligible for inclusion. Pathological nodal status could be either ypN0 (*i.e.*, pathologic complete response, pCR) or ypN+. Patients with known pathogenic germline *BRCA1/2* variants and those presenting with metastatic disease (cM+) at the initial diagnosis were excluded from the study. Data on age and menopausal status at diagnosis, tumor site, focality, dimensions at baseline ultrasound (US) and MRI, primary tumor stage, clinical nodal status, tumor grade, molecular features, and intrinsic subtype, type of NAT regimens, and NAT discontinuation were collected. Pre-operative staging involved bilateral mammogram, breast, and axillary US in the majority of patients. While pre- and post-operative MRI of the breast was not mandatory initially, its frequency increased during the study period as later internal guidelines recommended it for loco-regional staging post-NAT. The evaluation of hormone receptor status was conducted through immunohistochemistry (IHC). Estrogen (ER) and progesterone receptor (PgR) status was defined as positive when their expression was observed in at least 1 % of immunoreactive cells [21]. The assessment of human epidermal growth factor receptor 2 (HER2) status involved both IHC and fluorescent *in-situ* hybridization (FISH) techniques, following the ASCO-CAP guidelines [22]. To monitor the response to systemic treatment, patients underwent post-NAT evaluation using one or a combination of the following: bilateral breast and axillary US, MRI of the breast, or positron emission tomography-computed tomography (PET/CT). Management of each BC patient was discussed three times at multidisciplinary tumor board meetings, which included breast/oncoplastic surgeons, medical oncologists, pathologists, radiation oncologists, and radiologists. The first multidisciplinary tumor board meeting was held before initiating NAT, during which the decision was made to proceed with NAT followed by surgery; however, the type of breast surgery (BCS or mastectomy) was not specified at this time. The second multidisciplinary meeting took place after completion of NAT and before surgery to evaluate tumor response. The third and final multidisciplinary tumor board meeting was held after surgery to discuss pathological findings and plan adjuvant treatments.

2.2. Statistical analysis

Patients were divided into two groups according to the surgical procedure performed: BCS or mastectomy. Descriptive statistics were used for cohort characteristics. Continuous variables were summarized with median and interquartile range (IQR), and the Wilcoxon rank-sum test was employed for comparison. Binary and categorical variables were summarized using proportions. Two-sample tests of proportions and the Chi square test for heterogeneity were used for comparison, respectively. Univariate logistic regression was performed; variables with a significant correlation were then tested in a multivariate model. Distant recurrence-free survival (DRFS), defined as time from breast surgery to first occurrence of relapse in a distant site, and event-free survival (EFS), defined as time from breast surgery to first occurrence of relapse at any site or death, were evaluated. The Kaplan-Maier method was used for curves visualization and the log-rank test to define significant differences in survival between patients treated with BCS or mastectomy.

For all statistical tests, the significance level was set at 0.05. All analyses were conducted using STATA v.18 (StataCorp. 2023. Stata Statistical Software: Release 18. College Station, TX: StataCorp LLC).

3. Results

3.1. Population characteristics

Overall, 1407 BC patients were treated with NAT in the two breast units during the considered study period (2009–2022). Of these, 243 Caucasian women who achieved ypT0 were included, 147 (60.5 %) undergoing BCS and 96 (39.5 %) undergoing mastectomy. The distribution of surgical types was balanced across two hospital sites ($p = 0.87$). The median age was 50 years (IQR: 44–58), and menopausal status was similarly distributed ($p = 0.46$) between both groups. The upper external quadrant was the most common tumor site for both BCS and mastectomy patients. However, a notable prevalence of centrally-located cancers was observed in the mastectomy group (30.2 % vs 19.7 % in the BCS group, $p = 0.009$). Multiple foci on initial US were more frequent in the mastectomy group (33.3 % vs 16.4 % in the BCS group; $p = 0.001$), and tumors in this group were larger at baseline US (median extension 29 mm vs 24 mm in the BCS group; $p = 0.009$). Baseline MRI was performed in 63.4 % of patients, without significant differences between groups ($p = 0.97$). Mastectomy patients had more frequently multiple foci on MRI (42.7 % vs 27.2 % in BCS; $p = 0.005$) and larger tumor dimensions (median extension 40 mm vs 30 mm in BCS; $p = 0.0001$). The primary tumor stage (cT stage) showed a higher proportion of cT3/4 cancers in the mastectomy group (18.8 % vs 6.8 % in BCS; $p = 0.003$), leading to a greater number of stage III BC cases (14.6 % vs 6.1 % in BCS; $p = 0.03$). No significant differences were observed in tumor grading, hormone receptor status, HER2 status, Ki67 index, and intrinsic molecular subtype between groups. Chemotherapy regimens, primarily sequential anthracycline and taxane, was similar across groups ($p = 0.73$), as was the use of targeted agents (58.4 %, $p = 0.58$). Chemotherapy discontinuation rates were comparable (10.3 %, $p = 0.71$). Descriptive statistics are summarized in Table 1.

3.2. Regression analysis

In order to identify potential predictors of mastectomy in our population, we conducted univariate and multivariate logistic regression analyses. In the univariate analysis, several factors emerged as significantly related to the likelihood of receiving a mastectomy. Patients with multiple tumor foci at baseline US were more likely to undergo mastectomy, with an odds ratio (OR) of 2.69 [95 % confidence interval (95 % CI): 1.45–4.99, $p = 0.002$]. The cancer extension at baseline US was also a significant predictor, with each millimeter increase in tumor size associated with a 3 % increase in the likelihood of mastectomy (OR: 1.03, 95 % CI: 1.01–1.04, $p = 0.008$). Similarly, the tumor's focality and size at baseline MRI influenced the surgical decision: multiple foci were associated with a higher probability of mastectomy (OR: 2.59, 95 % CI: 1.33–5.04, $p = 0.005$), while each millimeter increase correlated with a 5 % higher chance of mastectomy (OR: 1.05, 95 % CI: 1.02–1.07, $p <$

Table 1
Descriptive statistics.

	Total N = 243	BCS N = 147 (60.5 %)	Mastectomy N = 96 (39.5 %)	p
Hospital site				0.87
Catania, n (%)	158 (65.0)	95 (64.6)	63 (65.6)	
Rozzano, n (%)	85 (35.0)	52 (35.4)	33 (34.4)	
Patients characteristics				
Age in years, median (IQR)	50 (44–58)	50 (45–58)	49.5 (43.5–47.5)	0.69
Menopausal status				0.46
Premenopausal, n (%)	122 (50.2)	71 (48.3)	51 (53.1)	
Postmenopausal, n (%)	121 (49.8)	76 (51.7)	45 (49.9)	
Tumor characteristics				
Site				
UpperExternal[a], n (%)	114 (46.9)	74 (50.3)	40 (41.7)	**0.009**
UpperInternal, n (%)	28 (11.5)	16 (11.0)	12 (12.5)	
Lower External, n (%)	29 (11.9)	20 (13.6)	9 (9.4)	
Lower Internal, n (%)	9 (3.7)	8 (5.4)	1 (1.0)	
Central, n (%)	58 (23.9)	29 (19.7)	29 (30.2)	
Multiple, n (%)	5 (2.1)	0 (0)	5 (5.2)	
Focality (baseline US)				**0.001**
Single, n (%)	172 (70.8)	115 (78.2)	57 (59.4)	
Multiple, n (%)	56 (23.0)	24 (16.4)	32 (33.3)	
Missing, n (%)	15 (6.2)	8 (5.4)	7 (7.3)	
Extension in mm (baseline US), median (IQR)	25 (18–35)	24 (17–30)	29 (20–40)	**0.009**
Baseline MRI, n (%)	154 (63.4)	93 (63.2)	61 (63.5)	0.97
Extension in mm (baseline MRI), median (IQR)	32 (23–45)	30 (21–36)	40 (29–60)	**0.0001**
Focality (baseline MRI)				**0.005**
Single	74 (30.5)	53 (36.1)	21 (21.9)	
Multiple	81 (33.3)	40 (27.2)	41 (42.7)	
Missing	88 (36.2)	54 (36.7)	34 (35.4)	
Post-treatment MRI, n (%)	165 (68)	104 (70.7)	61 (63.5)	0.24
Extension in mm (post-treatment MRI), median (IQR)	0 (0–7)	0 (0–6)	0 (0–11)	0.50
ycT0 at post-treatment MRI				0.69
Yes, n (%)	114 (46.9)	73 (49.7)	41 (42.7)	
No, n (%)	51 (21.0)	31 (21.1)	20 (20.8)	
Missing	78 (32.1)	43 (29.2)	35 (36.5)	
Primary tumor stage (cT)				**0.003**
cT1, n (%)	64 (26.3)	47 (32.0)	17 (17.7)	
cT2, n (%)	148 (61.0)	88 (59.9)	60 (62.5)	
cT3-4, n (%)	28 (11.5)	10 (6.8)	18 (18.8)	
Missing	3 (1.2)	2 (1.3)	1 (1.0)	
Clinical nodal stage (cN)				0.13
cN0, n (%)	102 (42.0)	67(45.6)	35 (36.5)	
cN+, n (%)	141 (58.0)	80 (54.4)	61 (63.5)	
Clinical TNM stage				**0.03**
I, n (%)	34 (14.0)	25 (17.0)	9 (9.4)	
II, n (%)	182 (74.9)	111 (75.5)	71 (73.9)	
III, n (%)				
Missing, n (%)	23 (9.5) 4 (1.6)	9 (6.1) 2 (1.4)	14 (14.6) 2 (2.1)	
Grading				0.61
Intermediate, n (%)	48 (19.7)	28 (19.1)	20 (20.8)	
High, n (%)	136 (56.0)	85 (57.8)	51 (53.1)	
Missing n (%)	59 (24.3)	34 (23.1)	25 (26.0)	
ER status				0.39
Positive, n (%)	116 (47.8)	74 (50.3)	42 (43.7)	
Negative, n (%)	125 (51.4)	73 (49.7)	52 (54.2)	
Missing, n (%)	2 (0.8)	0 (0)	2 (2.1)	
ER expression level, median (IQR)	0 (0–75)	1 (0–80)	0 (0–70)	0.49
PgR status				0.18
Positive, n (%)	92 (37.9)	61 (41.5)	31 (32.3)	
Negative, n (%)	149 (61.3)	86 (58.5)	63 (65.6)	
Missing, n (%)	2 (0.8)	0 (0)	2 (2.1)	
PgR expression level, median (IQR)	0 (0–20)	0 (0–25)	0 (0–15)	0.23
HER2 status				0.73
Positive, n (%)	140 (57.6)	86 (58.5)	54 (56.3)	
Negative, n (%)	103 (42.4)	61 (41.5)	42 (43.7)	
Ki67 % proliferation index, median (IQR)	40 (30–60)	40 (30–65)	40 (30–60)	0.29
Intrinsic variant				0.85
Luminal/HER2 neg, n (%)	45 (18.5)	28 (19.0)	17 (17.7)	
Luminal/HER2 pos, n (%)	71 (29.2)	46 (31.3)	25 (26.0)	
HER2 enriched, n (%)	67 (27.6)	39 (26.6)	28 (29.2)	
Triple negative, n (%)	58 (23.9)	34 (23.1)	24 (25.0)	
Missing, n (%)	2 (0.8)	0 (0)	2 (2.1)	
Treatment Characteristics				
Chemotherapy regimen, n (%)				0.73
Anthracycline + Taxane	222 (91.4)	136 (92.5)	86 (89.6)	
Anthracycline + Taxane + Carboplatin	12 (4.9)	7 (4.8)	5 (5.2)	
Anthracycline alone	3 (1.2)	1 (0.7)	2 (2.1)	
Taxane alone	6 (2.5)	3 (2.0)	3 (3.1)	
Targeted agents, n (%)	142 (58.4)	88 (59.9)	54 (56.2)	0.58
Chemotherapy discontinuation, n (%)	25 (10.3)	16 (10.9)	9 (9.4)	0.71

Legend: BCS breast-conserving surgery; IQR interquartile range; US ultrasound; MRI magnetic resonance imaging; ER estrogen receptor; PgRprogesterone receptor.

[a] Includes axillary tail.

0.0001). Patients with cT3-4 tumors had significantly higher odds of mastectomy compared to those with smaller tumors (OR: 5.52, 95 % CI: 1.91–16.01, p = 0.001). Additionally, patients with TNM stage III disease were more likely to undergo mastectomy compared to those with stages I or II (OR: 4.32, 95 % CI: 1.39–13.4, p = 0.01). Univariate analysis results are summarized in Table 2.

In a multivariate regression model that included variables with a statistically significant correlation in the univariate analysis (*i.e.*, baseline US focality, baseline US extension, baseline MRI focality, baseline MRI extension, primary tumor's stage, and TNM tumor's stage), only US focality and MRI extension retained their significance (OR 2.69, 95 % CI: 1.04–6.92; p = 0.041 and OR 1.04, 95 % CI: 1.01–1.09; p = 0.015, respectively, Pseudo R2 0.16) (Table 3).

3.3. Survival analysis

Median follow-up was 38.9 months in the BCS group (IQR 23.2–57.6 months) and 38.4 months in the mastectomy group (IQR 23.1–68.5 months).

Overall, 21 patients (8.6 %) had a distant recurrence, 8 (5.4 %) in the BCS group and 13 in the mastectomy group (13.5 %). At 12, 24 and 36 months DRFS was 96.5 %, 95.0 % and 94.0 % among patients treated with BCS and 95.7 %, 89.4 % and 86.4 % among patients treated with mastectomy, respectively. Distant-recurrence free survival difference was significant between the two groups (log-rank p = 0.031) (Fig. 1A).

Table 2
Univariate regression analysis.

Variables	Univariate odds ratio (95 % CI)	P value
Hospital site	0.96 (0.56–1.64)	0.87
Age in years	1.0 (0.97–1.02)	0.85
Menopausal status	0.82 (0.49–1.38)	0.46
Site		
Upperexternal[a]	–	–
Upperinternal	1.42 (0.61–3.31)	0.41
Lowerexternal	0.85 (0.36–2.05)	0.72
Lower internal	0.24 (0.03–1.97)	0.18
Central	1.90 (0.99–3.61)	0.05
Multiple	–	–
Focality (baselineUS)		
Single	–	–
Multiple	2.69 (1.45–4.99)	**0.002**
Extension in mm (baseline US)	1.03 (1.01–1.04)	**0.008**
Baseline MRI	1.01 (0.59–1.63)	0.97
Extension in mm (baseline MRI)	1.05 (1.02–1.07)	**<0.0001**
Focality (baseline MRI)		
Single	–	–
Multiple	2.59 (1.33–5.04)	**0.005**
Post-treatment MRI	0.72 (0.42–1.25)	0.24
Extension in mm (post-treatment MRI)	1.02 (0.99–1.05)	0.23
ycT0 at post-treatment MRI	0.87 (0.44–1.72)	0.69
Primary tumor stage (cT)		
cT1	–	–
cT2	1.88 (0.99–3.54)	0.054
cT3-4	4.98 (1.92–12.89)	**0.001**
Clinical nodal stage (cN) cN+	1.50 (0.88–2.55)	0.13
Clinical TNM stage		
I	–	–
II	1.77 (0.78–4.03)	0.17
III	4.32 (1.39-13-4)	**0.01**
Grading		
Intermediate	–	–
High	0.84 (0.43–1.64)	0.61
Estrogen receptor status	0.80 (0.47–1.34)	0.39
Estrogen receptor expression level	1.00 (0.99–1.00)	0.45
Progesterone receptor status	0.69 (0.40–1.19)	0.19
Progesterone receptor expression level	1.00 (0.99–1.00)	0.42
HER2 status	0.91 (0.54–1.53)	0.73
Ki67 % proliferation index	0.99 (0.98–1.00)	0.19
Intrinsic variant		
Luminal/HER2 neg	–	–
Luminal/HER2 pos	0.85 (0.39–1.84)	0.67
HER2 enriched	1.14 (0.53–2.48)	0.74
Triple negative	1.12 (0.34–2.50)	0.78
Chemotherapy regimen		
Anthracycline + Taxane	–	–
Anthracycline + Taxane + Carboplatin	1.13 (0.35–3.67)	0.84
Anthracycline alone	3.16 (0.28–35.4)	0.35
Taxane alone	1.58 (0.31–8.01)	0.58
Targeted agents	0.86 (0.51–1.45)	0.58
Chemotherapy discontinuation	0.85 (0.36–2.00)	0.71

Legend: US ultrasound; MRI magnetic resonance imaging; ER estrogen receptor; PgRprogesterone receptor.
[a] Includes axillary tail.

Twenty-seven patients (11.1 %) had an event, intended as loco-regional or distant recurrence or death for any cause, 11 (7.5 %) among BCS patients and 16 (16.7 %) among mastectomy patients. At 12, 24 and 36 months DRFS was 95.1 %, 93.6 % and 92.6 % among patients treated with BCS and 94.6 %, 87.0 % and 83.9 % among patients treated with mastectomy, respectively. Event-free survival difference was significant between the two groups (log-rank p = 0.034) (Fig. 1B).

4. Discussion

The use of NAT in BC has substantially increased over the past years

Table 3
Multivariate analysis.

Variables	Multivariate odds ratio (95 % CI)	P value
Focality (baseline US)		
Single	–	–
Multiple	2.69 (1.04–6.92)	**0.041**
Extension in mm (baseline US)	1.00 (0.95–1.04)	0.87
Focality (baseline MRI)		
Single	–	
Multiple	1.35 (0.55–3.31)	0.51
Extension in mm (baseline MRI)	1.04 (1.01–1.09)	**0.015**
Primary tumor stage (cT)		
cT1	–	–
cT2	1.59 (0.33–7.67)	0.57
cT3-4	–	-
Clinical TNM stage		
I	–	–
II	0.64 (0.10–4.07)	0.64
III	–	-

Legend: US ultrasound; MRI magnetic resonance imaging.

[13,23]. Pre-operative systemic treatment may lead to conversion from mastectomy to BCS, especially when breast conservation was not initially feasible. This is particularly true for BC patients who achieve complete remission of the primary tumor (*i.e.*, ypT0). Even though a considerable number of eligible patients undergo BCS following NAT, mastectomy is still performed in many cases. In recent neo-adjuvant trials, low rates of breast conservation have been observed despite the increase of pCR rates with innovative therapies [14,18,19,24–28]. A meta-analysis of 36 studies involving 12,311 BC patients confirmed that pCR does not increase BCS in patients receiving NAT [29].

We conducted a retrospective analysis of BC patients with ypT0 treated in two Italian Breast Units, to assess factors influencing surgical decision-making and identify predictors of mastectomy following NAT in this population. Data were extracted from a standardized, high-quality database, making our results comparable and widely applicable across the network that utilizes DataBreast.

According to regression analysis, cancer extension and focality on baseline US and MRI significantly correlated with the post-NAT surgical choice.

Pre-operative radiological assessment after NAT is more challenging compared to the upfront surgical setting. Tumor histopathological and biological features, heterogeneous response patterns, and potential inconsistencies between clinical and pathological responses can all contribute to the lack of confidence in post-NAT radiological evaluations [30–32]. Additionally, general consensus about the optimal radiological work-up after NAT has not been achieved yet [33–35]. Keune et al. conducted a retrospective analysis of 192 BC patients with 196 primary tumors and reported that US was more accurate than mammography in measuring residual disease after NAT. Ultrasound accurately sized the residual disease in 91.3 % of cases, compared to 51.9 % measured by mammography ($p < 0.0001$) [36]. Sudhir et al. compared post-NAT mammography, US, and MRI in 56 consecutive stage II-III BC patients [37]. Accuracy rates for predicting residual tumor size were 57 % for mammography, 50 % for US and 85 % for MRI. The sensitivity of MRI to detect pCR was 72.7 % [37]. A meta-analysis of 44 studies conducted between 1990 and 2008, involving 2050 BC patients who underwent imaging evaluation of residual disease after NAT, found that MRI exhibited generally high sensitivity (83–87 %) and variable specificity (54–83 %). However, no differences between MRI and US were found in terms of accuracy ($p = 0.15$) [38]. In a systematic review comprising 35 studies, the sensitivity, specificity, positive predictive value, and negative predictive value of MRI to predict pCR ranged from 25 % to 100 %, 50 %–97 %, 47 %–73 %, and 71 %–100 %, respectively [39]. A retrospective study by Kuzmova et al. demonstrated an overall accuracy of 77.3 % for MRI in predicting pCR, with significantly better performance in triple-negative BC compared to hormone

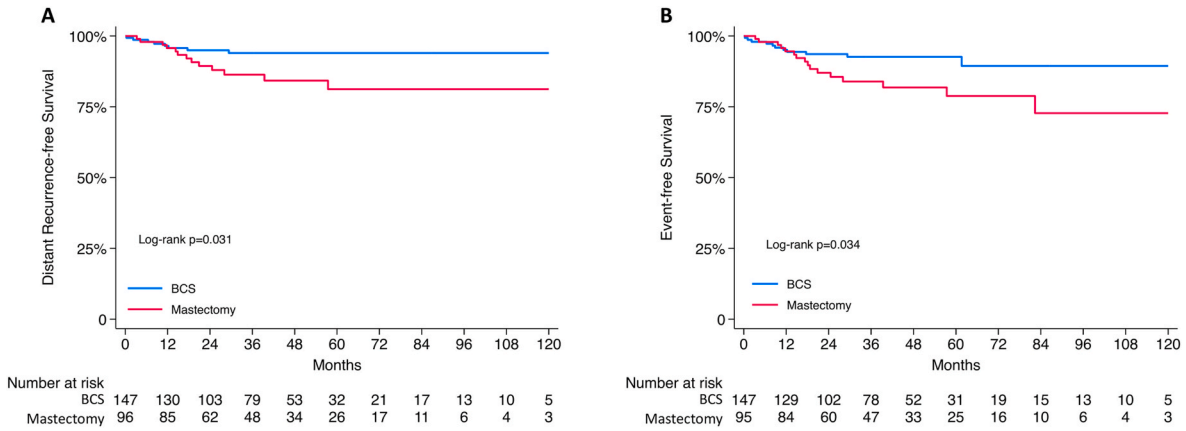


Fig. 1. Distant Recurrence-free Survival (A) and Event-free Survival (B) of ypT0 patients treated with breast-conserving surgery or mastectomy. BCS: Breast-conserving surgery.

receptor-positive/HER2-negative sub-type (p = 0.0057) [40]. Given its accuracy, MRI is considered the standard tool for primary tumor assessment after NAT [39,41,42]. However, it may overestimate the extent of residual disease, eventually leading to radical surgery and poorer cosmetic outcomes even when BCS is technically feasible and clinically appropriate [43–45]. Marinovich et al. [46] conducted a meta-analysis of 19 studies, examining the agreement between MRI and pathology in this setting. The study showed a tendency for MRI to overestimate pathologic tumor size, with a pooled mean difference of 0.1 cm. According to a retrospective study by Kwak et al. including 225 BC patients who underwent a MRI before and after NAT between 2016 and 2021, 15 % of the results were false positives [47].

Multiple tumor foci at baseline assessment was another factor significantly correlating with the choice of mastectomy in our study. Surgical decision-making in BC patients with more than two lesions in a single quadrant (*i.e.*multifocal) or across multiple quadrants (*i.e.*multicentric) often involves complex considerations and a degree of uncertainty for both patients and breast surgeons, especially after NAT. Guidelines usually indicate mastectomy for BC patients with unclear pre-operative margin status, including multifocal/centric presentations [48]. These recommendations are partly based on previous evidence showing poorer prognosis in multifocal/multicentric disease [49–51]. However, a recent pooled analysis of three prospective trials involving 6134 patients undergoing BCS after NAT showed that local control and OS are not inferior in BC patients with multicentric/multifocal tumors compared to those with unifocal presentation, provided that clear surgical margins are achieved [52].

Tumor shrinkage after NAT poses a challenge for breast surgeons that must accurately locating and completely excising the primary tumor site. Still, the use of metallic clip to mark the tumor bed in BC patients potentially candidate to BCS should be routing practice before NAT [53,54]. The metallic clip remains detectable during the entire course of NAT, facilitating pre-operative localization and intra-operative identification of the tumor bed [55].

In the upfront surgical setting, BCS has proven to be a safe alternative to mastectomy in terms of OS [56,57], with recent studies even reporting improved survival with BCS [58,59]. Similar results were obtained in patients undergoing BCS after NAT, showing that breast conservation is not detrimental compared to mastectomy, while providing better cosmetic outcomes and quality of life [20,60,61]. However, fear of local recurrence may represent another driver of mastectomy choice in patients with tumor complete response after NAT. This concern was accentuated by findings from a meta-analysis by the Early BC Trialists' Collaborative Group (EBCTCG) in 2018 [62], including ten randomized trials with a median follow-up of nine years, indicated a higher incidence of local recurrences in patients who underwent BCS for NAT-downsized tumors, compared to those receiving BCS in the adjuvant setting for similarly sized tumors. Notably, within these trials, two studies did not conduct surgical intervention post-NAT in cases of clinical complete response. The likelihood of local recurrence was particularly elevated in those initially designated for mastectomy but who ultimately received BCS, highlighting potential issues in surgical planning (*i.e.* radiological imaging and pre-operative tumor localization). Moreover, data on axillary surgery and radiotherapy were unknown. On the other hand, Werutsky et al. performed a pooled analysis including 10,075 patients with primary BC after NAT, showing that the 5-year cumulative loco-regional recurrence rates were 7.8 % among patients treated with BCS and 11.3 % with mastectomy [63]. In the recent prospective, randomized, I-SPY2 trial, Mukhtar et al. evaluated the relationship between the response to NAT, as measured by the residual cancer burden (RCB) method, and the loco-regional recurrence rate in 1462 BC patients who underwent surgical treatment (BCS vs mastectomy) between the years 2010 and 2021 [64]. At a median follow-up of 3.5 years, 34 (5.4 %) patients in the BCS group and 58 (7.0 %) patients in the mastectomy group experienced a loco-regional recurrence (p = 0.18). Those patients with RCB 2/3 disease after NAT experienced a significantly shorter loco-regional free survival than those with RCB 0/1 disease, regardless of whether BCS or mastectomy was performed. Among patients with RCB 0/1 status after NAT, there was no difference in loco-regional free survival between those having BCS vs mastectomy. Additionally, many other studies have consistently demonstrated that BCS does not compromise local recurrence rates or OS in BC patients treated with NAT [17–19]. Results of survival analysis in our cohort are in line with previous findings, demonstrating that BCS is not detrimental compared to mastectomy in patients responding to NAT [65–69]. Despite in our study DRFS and EFS are significantly better in patients undergoing BCS, these findings should be interpreted cautiously, due to the unbalance in baseline patients characteristics (e.g. more advanced stage in the mastectomy group) and lack of control for potential post-NAT biases (e.g. adjuvant treatments received).

Limitations of our study include its retrospective design, the lack of information about patient preferences and individual concerns, and the absence of post-NAT ultrasound data. However, we have implemented measures to mitigate them. To overcome the lack of post-NAT US data, we included post-NAT MRI assessments in our analysis, which provided comprehensive imaging information. By focusing on a homogeneous group of BC patients who achieved ypT0 after NAT and underwent surgery across two breast units, our study ensures a consistent treatment approach. The use of data from a standardized, high-quality database enhances the reliability of our findings. Although the study does not encompass considerations about oncoplastic procedure and delayed reconstruction, its structured approach and the specific patient cohort

enhance the generalizability and relevance of its results to other breast units.

Avoiding unnecessary mastectomies may enhance patient quality of life while safeguarding clinical outcomes. Multidisciplinary management and rigorous evaluation of patient and tumor characteristics before and after NAT is paramount for optimizing the treatment of these women. Our retrospective data suggest that mastectomy in patients with ypT0 disease post-NAT does not provide any advantage in terms of locoregional staging and long-term survival outcomes. Indeed, BCS may yield oncological outcomes that are comparable, if not superior, to those following mastectomy.

5. Conclusions

In conclusion, BCS should be prioritized whenever feasible to improve the management of BC patients responding to NAT. Prospective studies are needed to confirm our findings in the future.

Funding

The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

Competing interests

The authors have no relevant financial or non-financial interests to disclose that are directly or indirectly related to the work submitted for publication.

Data availability

The datasets generated during and/or analyzed during the current study are not publicly available due to patient confidentiality but are available from the corresponding author on reasonable request.

Ethics approval

No ethics approval is required for this study as this is a retrospective collection of data from a standard database for quality assurance assessment and not an observational study.

Consent to participate

Informed consent was obtained from all individual participants included in the study. Patients are consented for sharing data belonging to this database upon admission.

CRediT authorship contribution statement

Damiano Gentile: Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Visualization. **Federica Martorana:** Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Visualization. **Andreas Karakatsanis:** Conceptualization, Methodology, Investigation, Writing – original draft, Writing – review & editing, Supervision, Visualization. **Francesco Caruso:** Resources, Writing – review & editing. **Michele Caruso:** Resources, Writing – review & editing. **Gaetano Castiglione:** Resources, Writing – review & editing. **Alfio Di Grazia:** Resources, Writing – review & editing. **Francesco Pane:** Resources, Writing – review & editing. **Antonio Rizzo:** Resources, Writing – review & editing. **Paolo Vigneri:** Resources, Writing – review & editing. **Corrado Tinterri:** Resources, Writing – review & editing. **Giuseppe Catanuto:** Conceptualization, Methodology, Investigation, Resources, Writing – original draft, Writing – review & editing.

Declaration of competing interest

None

References

[1] Choudhary P, Gogia A, Deo S, et al. Neoadjuvant chemotherapy in locally advanced breast cancer: clinicopathological characteristics and correlation with pathological complete response. J Clin Oncol 2020;38:e12658. https://doi.org/10.1200/jco.2020.38.15_suppl.e12658.

[2] Wang M, Hou L, Chen M, et al. Neoadjuvant chemotherapy creates surgery opportunities for inoperable locally advanced breast cancer. Sci Rep 2017;7:44673. https://doi.org/10.1038/srep44673.

[3] Hortobagyi GN, Ames FC, Buzdar AU, et al. Management of stage III primary breast cancer with primary chemotherapy, surgery, and radiation therapy. Cancer 1988; 62:2507–16. https://doi.org/10.1002/1097-0142(19881215)62:12<2507::AID-CNCR2820621210>3.0.CO;2-D.

[4] Cortazar P, Zhang L, Untch M, et al. Pathological complete response and long-term clinical benefit in breast cancer: the CTNeoBC pooled analysis. Lancet 2014;384: 164–72. https://doi.org/10.1016/S0140-6736(13)62422-8.

[5] Von Minckwitz G, Blohmer JU, Costa SD, et al. Response-guided neoadjuvant chemotherapy for breast cancer. J Clin Oncol 2013;31:3623–30. https://doi.org/10.1200/JCO.2012.45.0940.

[6] Gentile D, Sagona A, De Carlo C, et al. Pathologic response and residual tumor cellularity after neo-adjuvant chemotherapy predict prognosis in breast cancer patients. Breast 2023;69:323–9. https://doi.org/10.1016/j.breast.2023.03.016.

[7] Von Minckwitz G, Untch M, Blohmer JU, et al. Definition and impact of pathologic complete response on prognosis after neoadjuvant chemotherapy in various intrinsic breast cancer subtypes. J Clin Oncol 2012;30:1796–804. https://doi.org/10.1200/JCO.2011.38.8595.

[8] Yoshioka T, Hosoda M, Yamamoto M, et al. Prognostic significance of pathologic complete response and Ki67 expression after neoadjuvant chemotherapy in breast cancer. Breast Cancer 2015;22:185–91. https://doi.org/10.1007/s12282-013-0474-2.

[9] Broglio KR, Quintana M, Foster M, et al. Association of pathologic complete response to neoadjuvant therapy in HER2-positive breast cancer with long-term outcomes ameta-analysis. JAMA Oncol 2016;2:751–60. https://doi.org/10.1001/jamaoncol.2015.6113.

[10] Tinterri C, Sagona A, Barbieri E, et al. Sentinel lymph node biopsy in breast cancer patients undergoing neo-adjuvant chemotherapy: clinical experience with node-negative and node-positive disease prior to systemic therapy. Cancers 2023;15: 1719. https://doi.org/10.3390/cancers15061719.

[11] Boughey JC, Yu H, Dugan CL, et al. Changes in surgical management of the axilla over 11 Years – report on more than 1500 breast cancer patients treated with neoadjuvant chemotherapy on the prospective I-SPY2 trial. Ann Surg Oncol 2023; 30:6401–10. https://doi.org/10.1245/s10434-023-13759-y.

[12] Clough KB, Acosta-Marín V, Nos C, et al. Rates of neoadjuvant chemotherapy and oncoplastic surgery for breast cancer surgery: a French national survey. Ann Surg Oncol 2015;22:3504–11. https://doi.org/10.1245/s10434-015-4378-6.

[13] Vugts G, Maaskant-Braat AJG, Nieuwenhuijzen GAP, et al. Patterns of care in the administration of neo-adjuvant chemotherapy for breast cancer. A population-based study. Breast J 2016;22:316–21. https://doi.org/10.1111/tbj.12568.

[14] Van der Hage JA, Van de Velde CJH, Julien JP, et al. Preoperative chemotherapy in primary operable breast cancer: results from the European organization for Research and treatment of cancer trial 10902. J Clin Oncol 2001;19:4224–37. https://doi.org/10.1200/JCO.2001.19.22.4224.

[15] Golshan M, Cirrincione CT, Sikov WM, et al. Impact of neoadjuvant chemotherapy in stage II-III triple negative breast cancer on eligibility for breast-conserving surgery and breast conservation rates: surgical results from CALGB 40603 (Alliance). Ann Surg 2015;262:434–8. https://doi.org/10.1097/SLA.0000000000001417.

[16] Wolmark N, Wang J, Mamounas E, et al. Preoperative chemotherapy in patients with operable breast cancer: nine-year results from National Surgical Adjuvant Breast and Bowel Project B-18. J Natl Cancer Inst Monogr 2001:96–102. https://doi.org/10.1093/oxfordjournals.jncimonographs.a003469.

[17] Gwark S, Kim HJ, Kim J, et al. Survival after breast-conserving surgery compared with that after mastectomy in breast cancer patients receiving neoadjuvant chemotherapy. Ann Surg Oncol 2023;30:2845–53. https://doi.org/10.1245/s10434-022-12993-0.

[18] Simons JM, Jacobs JG, Roijers JP, et al. Disease-free and overall survival after neoadjuvant chemotherapy in breast cancer: breast-conserving surgery compared to mastectomy in a large single-centre cohort study. Breast Cancer Res Treat 2021; 185:441–51. https://doi.org/10.1007/s10549-020-05966-y.

[19] Agrawal SK, Patel D, Shenoy P, et al. Oncologic safety of breast conservation following NACT in women with locally advanced breast cancer. Ecancermedicalscience 2023;17. https://doi.org/10.3332/ecancer.2023.1554.

[20] Pfob A, Dubsky P. The underused potential of breast conserving therapy after neoadjuvant system treatment – causes and solutions. Breast 2023;67:110–5. https://doi.org/10.1016/j.breast.2023.01.008.

[21] Allison KH, Hammond MEH, Dowsett M, et al. Estrogen and progesterone receptor testing in breast cancer: ASCO/CAP guideline update. J Clin Oncol 2020;38: 1346–66. https://doi.org/10.1200/JCO.19.02309.

[22] Wolff AC, Hammond MEH, Hicks DG, et al. Recommendations for human epidermal growth factor receptor 2 testing in breast. J Clin Oncol 2013;31:3997–4013. https://doi.org/10.1200/JCO.2013.50.9984.
[23] Spronk PER, Volders JH, van den Tol P, et al. Breast conserving therapy after neoadjuvant chemotherapy; data from the Dutch Breast Cancer Audit. Eur J Surg Oncol 2019;45:110–7. https://doi.org/10.1016/j.ejso.2018.09.027.
[24] Caudle AS, Gonzalez-Angulo AM, Hunt KK, et al. Impact of progression during neoadjuvant chemotherapy on surgical management of breast cancer. Ann Surg Oncol 2011;18:932–8. https://doi.org/10.1245/s10434-010-1390-8.
[25] Cristofanilli M, Gonzalez-Angulo A, Sneige N, et al. Invasive lobular carcinoma classic type: response to primary chemotherapy and survival outcomes. J Clin Oncol 2005;23:41–8. https://doi.org/10.1200/JCO.2005.03.111.
[26] Gusic LH, Walsh K, Flippo-Morton T, et al. Rationale for mastectomy after neoadjuvant chemotherapy. Am Surg 2018;84:126–32.
[27] Zarba Meli E, De Santis A, Cortese G, et al. Nipple-sparing mastectomy after neoadjuvant chemotherapy: definitive results with a long-term follow-up evaluation. Ann Surg Oncol 2023;30:2163–72. https://doi.org/10.1245/s10434-022-13035-5.
[28] Criscitiello C, Curigliano G, Burstein HJ, et al. Breast conservation following neoadjuvant therapy for breast cancer in the modern era: are we losing the opportunity? Eur J Surg Oncol 2016;42:1780–6. https://doi.org/10.1016/j.ejso.2016.10.011.
[29] Criscitiello C, Golshan M, Barry WT, et al. Impact of neoadjuvant chemotherapy and pathological complete response on eligibility for breast-conserving surgery in patients with early breast cancer: a meta-analysis. Eur J Cancer 2018;97:1–6. https://doi.org/10.1016/j.ejca.2018.03.023.
[30] Fei J, Wang GQ, Meng YY, et al. Breast cancer subtypes affect the ultrasound performance for axillary lymph node status evaluation after neoadjuvant chemotherapy: a retrospective analysis. Jpn J Clin Oncol 2021;51:1509–14. https://doi.org/10.1093/jjco/hyab117.
[31] Baumgartner A, Tausch C, Hosch S, et al. Ultrasound-based prediction of pathologic response to neoadjuvant chemotherapy in breast cancer patients. Breast 2018;39:19–23. https://doi.org/10.1016/j.breast.2018.02.028.
[32] Tinterri C, Fernandes B, Zambelli A, et al. The impact of different patterns of residual disease on long-term oncological outcomes in breast cancer patients treated with neo-adjuvant chemotherapy. Cancers 2024;16:376. https://doi.org/10.3390/cancers16020376.
[33] Sperber F, Weinstein Y, Sarid D, et al. Preoperative clinical, mammographic and sonographic assessment of neoadjuvant chemotherapy response in breast cancer. Isr Med Assoc J 2006;8:342–6.
[34] Loehberg CR, Lux MP, Ackermann S, et al. Neoadjuvant chemotherapy in breast cancer: which diagnostic procedures can be used? Anticancer Res 2005;25:2519–25.
[35] Dialani V, Chadashvili T, Slanetz PJ. Role of imaging in neoadjuvant therapy for breast cancer. Ann Surg Oncol 2015;22:1416–24. https://doi.org/10.1245/s10434-015-4403-9.
[36] Keune JD, Jeffe DB, Schootman M, et al. Accuracy of ultrasonography and mammography in predicting pathologic response after neoadjuvant chemotherapy for breast cancer. Am J Surg 2010;199:477–84. https://doi.org/10.1016/j.amjsurg.2009.03.012.
[37] Sudhir R, Koppula VC, Rao TS, et al. Accuracy of digital mammography, ultrasound and MRI in predicting the pathological complete response and residual tumor size of breast cancer after completion of neoadjuvant chemotherapy. Indian J Cancer 2022;59:345–53. https://doi.org/10.4103/ijc.IJC_795_19.
[38] Marinovich ML, Houssami N, Macaskill P, et al. Meta-analysis of magnetic resonance imaging in detecting residual breast cancer after neoadjuvant therapy, vol. 105. JNCI J Natl Cancer Inst; 2013. p. 321–33. https://doi.org/10.1093/jnci/djs528.
[39] Lobbes MBI, Prevos R, Smidt M, et al. The role of magnetic resonance imaging in assessing residual disease and pathologic complete response in breast cancer patients receiving neoadjuvant chemotherapy: a systematic review. Insights Imaging 2013;4:163–75. https://doi.org/10.1007/s13244-013-0219-y.
[40] Kuzmova M, Cullinane C, Rutherford C, et al. The accuracy of MRI in detecting pathological complete response following neoadjuvant chemotherapy in different breast cancer subtypes. Surg Oncol 2023;51:102011. https://doi.org/10.1016/j.suronc.2023.102011.
[41] Yuan Y, Chen X-S, Liu S-Y, Shen K-W. Accuracy of MRI in prediction of pathologic complete remission in breast cancer after preoperative therapy: a meta-analysis. Am J Roentgenol 2010;195:260–8. https://doi.org/10.2214/AJR.09.3908.
[42] Wu L-M, Hu J-N, Gu H-Y, et al. Can diffusion-weighted MR imaging and contrast-enhanced MR imaging precisely evaluate and predict pathological response to neoadjuvant chemotherapy in patients with breast cancer? Breast Cancer Res Treat 2012;135:17–28. https://doi.org/10.1007/s10549-012-2033-5.
[43] Belli P, Costantini M, Malaspina C, et al. MRI accuracy in residual disease evaluation in breast cancer patients treated with neoadjuvant chemotherapy. Clin Radiol 2006;61:946–53. https://doi.org/10.1016/j.crad.2006.07.004.
[44] Sener SF, Sargent RE, Lee C, et al. MRI does not predict pathologic complete response after neoadjuvant chemotherapy for breast cancer. J Surg Oncol 2019;120:903–10. https://doi.org/10.1002/jso.25663.
[45] Marinovich ML, Macaskill P, Irwig L, et al. Agreement between MRI and pathologic breast tumor size after neoadjuvant chemotherapy, and comparison with alternative tests: individual patient data meta-analysis. BMC Cancer 2015;15:662. https://doi.org/10.1186/s12885-015-1664-4.
[46] Marinovich ML, Macaskill P, Irwig L, et al. Meta-analysis of agreement between MRI and pathologic breast tumour size after neoadjuvant chemotherapy. Br J Cancer 2013;109:1528–36. https://doi.org/10.1038/bjc.2013.473.
[47] Kwak L, Santa-Maria C, Di Carlo P, et al. Can breast MRI predict pathologic response following neoadjuvant chemotherapy for breast cancer? A retrospective cohort study. Clin Imag 2023;101:105–12. https://doi.org/10.1016/j.clinimag.2023.06.007.
[48] Gradishar WJ, Moran MS, Abraham J, et al. NCCN Guidelines® insights: breast cancer, version 4.2023. J Natl Compr Cancer Netw 2023;21:594–608. https://doi.org/10.6004/jnccn.2023.0031.
[49] Lynch SP, Lei X, Chavez-MacGregor M, et al. Multifocality and multicentricity in breast cancer and survival outcomes. Ann Oncol 2012;23:3063–9. https://doi.org/10.1093/annonc/mds136.
[50] Vera-Badillo FE, Napoleone M, Ocana A, et al. Effect of multifocality and multicentricity on outcome in early stage breast cancer: a systematic review and meta-analysis. Breast Cancer Res Treat 2014;146:235–44. https://doi.org/10.1007/s10549-014-3018-3.
[51] Wolters R, Wöckel A, Janni W, et al. Comparing the outcome between multicentric and multifocal breast cancer: what is the impact on survival, and is there a role for guideline-adherent adjuvant therapy? A retrospective multicenter cohort study of 8,935 patients. Breast Cancer Res Treat 2013;142:579–90. https://doi.org/10.1007/s10549-013-2772-y.
[52] Ataseven B, Lederer B, Blohmer JU, et al. Impact of multifocal or multicentric disease on surgery and locoregional, distant and overall survival of 6,134 breast cancer patients treated with neoadjuvant chemotherapy. Ann Surg Oncol 2015;22:1118–27. https://doi.org/10.1245/s10434-014-4122-7.
[53] Shahzad R, Masood M, Shahid A, et al. Sonographically guided metalic clip placement for tumour localization in early breast cancer patients undergoing neoadjuvant chemotherapy. J Pakistan Med Assoc 2019;69:1501–4.
[54] Youn I, Choi SH, Kook SH, et al. Ultrasonography-Guided surgical clip placement for tumor localization in patients undergoing neoadjuvant chemotherapy for breast cancer. J Breast Cancer 2015;18:44. https://doi.org/10.4048/jbc.2015.18.1.44.
[55] Minella C, Villasco A, D'Alonzo M, et al. Surgery after neoadjuvant chemotherapy: a clip-based technique to improve surgical outcomes, a single-center experience. Cancers 2022;14:2229. https://doi.org/10.3390/cancers14092229.
[56] Fisher B, Anderson S, Bryant J, et al. Twenty-year follow-up of a randomized trial comparing total mastectomy, lumpectomy, and lumpectomy plus irradiation for the treatment of invasive breast cancer. N Engl J Med 2002;347:1233–41. https://doi.org/10.1056/NEJMoa022152.
[57] Veronesi U, Cascinelli N, Mariani L, et al. Twenty-year follow-up of a randomized study comparing breast-conserving surgery with radical mastectomy for early breast cancer. N Engl J Med 2002;347:1227–32. https://doi.org/10.1056/NEJMoa020989.
[58] Agarwal S, Pappas L, Neumayer L, et al. Effect of breast conservation therapy vs mastectomy on disease-specific survival for early-stage breast cancer. JAMA Surg 2014;149:267. https://doi.org/10.1001/jamasurg.2013.3049.
[59] De la Cruz Ku G, Karamchandani M, Chambergo-Michilot D, et al. Does breast-conserving surgery with radiotherapy have a better survival than mastectomy? A meta-analysis of more than 1,500,000 patients. Ann Surg Oncol 2022;29:6163–88. https://doi.org/10.1245/s10434-022-12133-8.
[60] Matuschek C, Nestle-Kraemling C, Haussmann J, et al. Long-term cosmetic outcome after preoperative radio-/chemotherapy in locally advanced breast cancer patients. Strahlenther Onkol 2019;195:615–28. https://doi.org/10.1007/s00066-019-01473-2.
[61] Caudle AS, Kuerer HM. Breast conservation therapy after neoadjuvant chemotherapy: optimization of a multimodality approach. J Surg Oncol 2014;110:32–6. https://doi.org/10.1002/jso.23595.
[62] Asselain B, Barlow W, Bartlett J, et al. Long-term outcomes for neoadjuvant versus adjuvant chemotherapy in early breast cancer: meta-analysis of individual patient data from ten randomised trials. Lancet Oncol 2018;19:27–39. https://doi.org/10.1016/S1470-2045(17)30777-5.
[63] Werutsky G, Untch M, Hanusch C, et al. Locoregional recurrence risk after neoadjuvant chemotherapy: a pooled analysis of nine prospective neoadjuvant breast cancer trials. Eur J Cancer 2020;130:92–101. https://doi.org/10.1016/j.ejca.2020.02.015.
[64] Mukhtar RA, Chau H, Woriax H, et al. Breast conservation surgery and mastectomy have similar locoregional recurrence after neoadjuvant chemotherapy. Ann Surg 2023;278:320–7. https://doi.org/10.1097/SLA.0000000000005968.
[65] Arlow RL, Paddock LE, Niu X, et al. Breast-conservation therapy after neoadjuvant chemotherapy does not compromise 10-year breast cancer–specific mortality. Am J Clin Oncol 2018;41:1246–51. https://doi.org/10.1097/COC.0000000000000456.
[66] Levy A, Borget I, Bahri M, et al. Loco-regional control after neo-adjuvant chemotherapy and conservative treatment for locally advanced breast cancer patients. Breast J 2014;20:381–7. https://doi.org/10.1111/tbj.12277.
[67] Cho JH, Park JM, Park HS, et al. Oncologic safety of breast-conserving surgery compared to mastectomy in patients receiving neoadjuvant chemotherapy for locally advanced breast cancer. J Surg Oncol 2013;108:531–6. https://doi.org/10.1002/jso.23439.
[68] Carrara GFA, Scapulatempo-Neto C, Abrahão-Machado LF, et al. Breast-conserving surgery in locally advanced breast cancer submitted to neoadjuvant chemotherapy. Safety and effectiveness based on ipsilateral breast tumor recurrence and long-term follow-up. Clinics 2017;72:134–42. https://doi.org/10.6061/clinics/2017(03)02.
[69] Shin H-C, Han W, Moon H-G, et al. Breast-conserving surgery after tumor downstaging by neoadjuvant chemotherapy is oncologically safe for stage III breast cancer patients. Ann Surg Oncol 2013;20:2582–9. https://doi.org/10.1245/s10434-013-2909-6.